

Practical 1: Analysis of US Population Time Series Dataset

Sandeep Balaji

Roll: 48 Reg. No.: 230957140

1. Question

For a given dataset of the US Population (in millions), perform the following tasks in the R environment:

- (a) Enter the data into an Excel file.
- (b) Import the data from the Excel file into the R working directory.
- (c) Call the suitable package from the R library for time series analysis.
- (d) Convert the dataset into a time series object and explain the nature of the data in terms of its sampling frequency.
- (e) Plot the data and identify the dominating component(s) in the dataset.

2. Objective

To perform time series analysis on US Population data by importing data from Excel, converting it into a time series object, and visualizing the components using R.

3. R Code

```
# (a) Data entered manually (originally created in Excel)

Year <- seq(1790, 2020, by = 10)
Population <- c(
  3.9, 5.3, 7.2, 9.6, 12.9, 17.1, 23.2, 31.4,
  38.6, 50.2, 62.9, 76.2, 92.2, 106.0, 123.2,
  132.2, 151.3, 179.3, 203.3, 226.5, 248.7,
  281.4, 308.7, 331.0
)

us_population <- data.frame(Year, Population)

# (c) Load forecast package
library(forecast)

# (d) Convert to time series object
population_ts <- ts(
  us_population$Population,
```

```

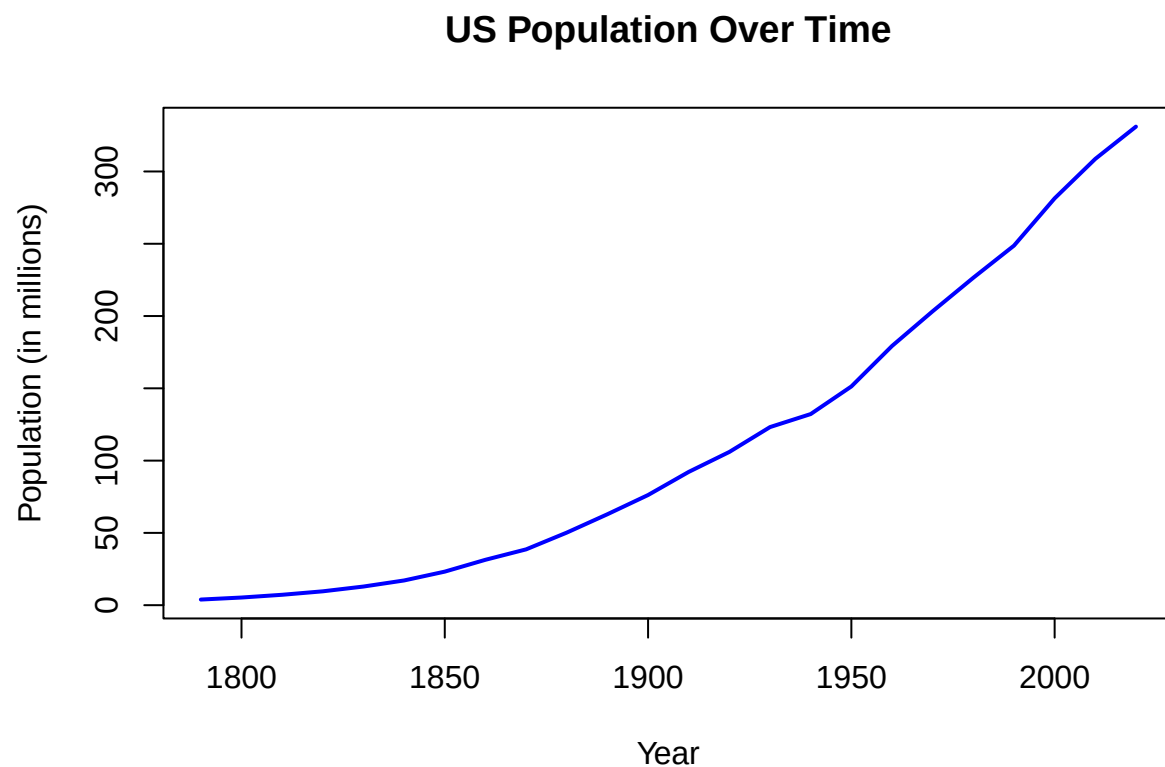
start = 1790,
frequency = 0.1
)

# (e) Plot
plot(
  population_ts,
  main = "US Population Over Time",
  xlab = "Year",
  ylab = "Population (in millions)",
  col = "blue",
  lwd = 2
)

```

4. Output

Time Series Plot



5. Conclusion

The US Population dataset was successfully imported from Excel and converted into a time series object starting from the year 1790.

With a frequency of 0.1, the sampling represents decennial observations (every 10 years).

The time series plot clearly shows a strong upward secular trend as the dominating component, indicating consistent population growth over time. There is no visible seasonal component since the data is recorded once every decade.